Noise-Robust Adaptive Meta-Adversarial Imitation Learning

IEEE Publication Technology, *Staff, IEEE,*

***Abstract*—This article proposes a noise-robust adaptive meta-adversarial imitation learning (AMAIL) approach to address key challenges in imitation learning, including training instability, low sample efficiency, and limited robustness under imperfect demonstrations. Specifically, a meta-learning network is introduced to extract structured knowledge from offline expert data and to provide intrinsic guidance for policy optimization. In addition, an expert confidence estimation module is developed to dynamically regulate the influence of imitation signals, enabling a balanced interaction between imitation learning and reinforcement learning during training. By incorporating meta-learning into a diffusion-based adversarial framework, the proposed method captures structural patterns in expert behavior while reducing the computational burden typically associated with diffusion models. A dynamic confidence weighting mechanism further enhances robustness by suppressing the influence of noisy or suboptimal demonstrations and by mitigating overfitting. Consequently, the proposed framework improves training stability and data efficiency. Extensive experiments on Gym MuJoCo and PyBullet demonstrate that the proposed method achieves faster convergence, improved stability, and competitive performance compared with representative baseline methods.**

***Index Terms*—Adversarial imitation learning, diffusion models, meta-learning, robust policy learning, robotic control.**

REFERENCES

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